

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – CLASS TEST (5 Questions)

Course Code: ITA05	Course Title: COMPUTER VISION
CO Assessed: CO2 – Manipulation (BL3)	Assessment: Assessment Tool 3 – Class Test
Weightage: 20% of CIA	Total Marks: 25 (5 Questions × 5 Marks)

CO5 Statement: Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BL3)

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Section / Batch: Slot C / 2023	Date: 22 / 08/26

Code of Conduct

I (DK Hari Krishna / 192321176) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: DK Hari Krishna

Instructions

- This test contains 5 questions, each carrying 10 marks. Total: 50 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

Section / Topic Focus	Q Nos	Marks
Image Enhancement in Spatial Domain Improving image quality using spatial domain techniques	1	10
Histogram Processing Techniques Enhancing image contrast through histogram equalization, histogram specification, and histogram stretching.	2	10
Spatial Filtering Noise reduction and feature enhancement	3	10
Image Enhancement in Frequency Domain Enhancing images using frequency domain transformations	4	10
Image Segmentation Partitioning images into meaningful regions	5	10

GRAND TOTAL: 50 Marks

Annexure A — Class Test: Answers

Q1. Image Enhancement in the Spatial Domain

Concept

Spatial domain enhancement refers to techniques that operate directly on the pixel values of an image, $g(x,y) = T[f(x,y)]$, where f is the input image, g is the output, and T is an operator defined over a neighborhood of (x,y) (a single pixel for point operations, or a local window for neighborhood/filtering operations). Unlike frequency-domain methods, spatial techniques manipulate intensity values directly, making them computationally simple and intuitive to apply.

Basic Grey-Level Transformations

These are point operations applied independently to each pixel, based only on its own intensity value:

- **Negative transformation:** reverses intensity levels ($s = L-1-r$), useful for highlighting white/grey detail embedded in dark regions such as medical images.
- **Log transformation:** ($s = c \cdot \log(1+r)$) compresses the dynamic range, expanding dark pixel values while compressing bright ones — useful for displaying images with a very large range of intensities (e.g., Fourier spectra).
- **Power-law (gamma) transformation:** ($s = c \cdot r^\gamma$) allows selective enhancement of either dark or bright regions depending on γ , and is widely used for gamma correction of display devices.
- **Contrast stretching:** linearly expands a narrow intensity range to the full display range, directly improving the visual contrast of low-contrast images.

Histogram Processing Techniques

Histogram-based methods use the global intensity distribution of the image rather than single pixels. Histogram equalization redistributes intensities to produce an approximately uniform histogram, automatically improving overall contrast. Histogram specification (matching) transforms the histogram of the input image to match a desired target histogram, giving more control than plain equalization. Histogram stretching linearly rescales the observed intensity range to occupy the full available range, similar in effect to contrast stretching but derived directly from histogram statistics.

Spatial Filtering

Spatial filtering uses a small neighborhood (mask/kernel) around each pixel rather than the pixel alone. Smoothing filters (mean, median, Gaussian) average or reorder neighborhood values to suppress noise and blur fine detail. Sharpening filters (Laplacian, unsharp masking, high-boost) accentuate rapid intensity changes to enhance edges and fine structures that were softened by blur or poor focus.

Justification of Suitability

- **Grey-level transformations** are the right first step whenever contrast or dynamic-range issues stem from the overall intensity distribution, since they need no spatial neighborhood information and are computationally cheap.
- **Histogram techniques** are best suited when the enhancement goal concerns overall image contrast (e.g., correcting under/over-exposed images) rather than a specific known reference range.
- **Spatial filtering** are essential when the degradation is spatially local — random noise, blurred edges, or loss of fine detail — since only a neighborhood operation can distinguish signal structure from isolated noise.

In practice, these techniques are often combined in sequence — e.g., contrast stretching or histogram equalization followed by noise-reducing smoothing and, where needed, sharpening — to obtain a well-balanced, visually improved image.

Q2. Image Smoothing vs. Image Sharpening

Differentiation

Image smoothing and image sharpening are complementary spatial-filtering operations that act on opposite ends of the intensity-variation spectrum. Smoothing (low-pass filtering) reduces the rate of change of intensity across the image — it blurs sharp transitions to remove noise and small unwanted variations. Sharpening (high-pass filtering) increases the rate of change of intensity — it accentuates edges, boundaries, and fine details that smoothing would otherwise suppress.

Aspect	Image Smoothing	Image Sharpening
Goal	Reduce noise / suppress rapid intensity variations	Enhance edges / accentuate rapid intensity variations
Underlying operation	Low-pass spatial filtering (averaging / weighted averaging)	High-pass spatial filtering (differentiation-based)
Effect on edges	Blurs edges and fine detail	Sharpens and highlights edges and fine detail
Effect on noise	Suppresses random noise effectively	Can amplify noise (since noise is high-frequency)
Typical operators	Mean filter, Median filter, Gaussian filter	Laplacian filter, Unsharp masking, High-boost filter
When to use	Noisy images, pre-processing before segmentation	Blurred images, enhancing fine structural detail

Working Principles

- **Mean filter:** replaces each pixel with the average of its neighborhood, reducing variance but blurring edges proportionally to the kernel size.
- **Median filter:** replaces each pixel with the median of its neighborhood; highly effective against impulsive (salt-and-pepper) noise while preserving edges better than the mean filter.
- **Gaussian filter:** uses distance-weighted averaging (heavier weight near the center) to smooth the image while producing more natural results than uniform averaging.
- **Laplacian filter:** computes the second derivative of intensity and adds it back to the original image, boosting regions of rapid intensity change (edges) while leaving flat regions unaffected.
- **Unsharp masking:** subtracts a blurred (smoothed) version of the image from the original to obtain a 'mask' of high-frequency detail, which is then added back to the original to enhance sharpness.

Improving Image Quality and Application Scenarios

Smoothing filters improve quality by removing noise introduced during acquisition or transmission, and are typically applied as a pre-processing step before edge detection or segmentation, where noise would otherwise cause false edges. Sharpening filters improve quality by restoring or emphasizing detail lost to defocus, motion blur, or previous smoothing, and are applied in the later stages of enhancement — for example, to make text or fine structural detail more legible, or to highlight subtle boundaries before feature extraction. The two are frequently used together: smoothing first to control noise, followed by controlled sharpening to recover perceptual detail without reintroducing the noise that was just removed.

Q3. Image Enhancement in the Frequency Domain

Concept

Frequency-domain enhancement transforms the image from the spatial domain into the frequency domain using the 2-D Fourier Transform, $F(u,v)$. Here, low frequencies correspond to slowly varying intensity regions (smooth backgrounds), while high frequencies correspond to rapid intensity changes (edges, noise, fine texture). Enhancement is performed by multiplying $F(u,v)$ with a filter function $H(u,v)$ that selectively attenuates or amplifies specific frequency components, and then applying the inverse Fourier Transform to obtain the enhanced spatial-domain image: $g(x,y) = F^{-1}[H(u,v) \cdot F(u,v)]$.

Roles of Low-Pass and High-Pass Filters

A low-pass filter (LPF) retains low-frequency components and attenuates high-frequency components, producing a smoothed, blurred output — this suppresses random noise (which is primarily high-frequency) at the cost of some fine detail. A high-pass filter (HPF) retains high-frequency components and attenuates low-frequency ones, emphasizing edges, boundaries, and fine texture while suppressing the smooth background — this sharpens the image and highlights structural detail.

Aspect	Low-Pass Filter (LPF)	High-Pass Filter (HPF)
Frequencies passed	Low frequencies (smooth, slowly varying regions)	High frequencies (edges, fine detail, noise)
Frequencies attenuated	High frequencies	Low frequencies
Visual effect	Blurring / smoothing of the image	Edge enhancement / sharpening
Common types	Ideal LPF, Butterworth LPF, Gaussian LPF	Ideal HPF, Butterworth HPF, Gaussian HPF
Typical application	Noise removal, image smoothing, data compression pre-processing	Edge detection, sharpening, fine-detail enhancement

Comparison of Effects and Suitable Applications

- **Low-pass filtering** smooths the image and reduces noise but can blur important edges if the cutoff frequency is set too low; suitable for pre-processing noisy images, softening block artifacts, or preparing images for compression.
- **High-pass filtering** sharpens edges and reveals fine structural detail but tends to amplify noise (since noise is also high-frequency) and can suppress overall brightness/contrast information carried by low frequencies; suitable for edge enhancement, feature extraction, and detecting fine defects or textures.

In practice, the choice between LPF and HPF (or a combination such as high-boost filtering, which selectively amplifies high frequencies while retaining some low-frequency content) depends on whether the primary goal is noise suppression (favoring LPF) or detail/edge enhancement (favoring HPF).

Q4. Image Segmentation

Concept

Image segmentation is the process of partitioning a digital image into multiple meaningful regions or sets of pixels that share common characteristics (intensity, texture, or discontinuity) so that objects of interest can be separated from the background or from each other. It is a critical intermediate step between low-level image processing and high-level image analysis/recognition.

Comparison of Segmentation Techniques

Technique	Working Principle	Advantages	Limitations
Point detection	Flags isolated pixels whose intensity differs sharply from their surroundings (Laplacian-type mask + threshold)	Simple; detects isolated bright/dark spots effectively	Only detects single-pixel anomalies; not useful for extended objects
Line detection	Uses directional masks (0° , 45° , 90° , 135°) to respond to thin line-like structures	Detects linear structures regardless of orientation	Sensitive to noise; limited to thin, straight features
Edge detection	Computes intensity gradients (Sobel, Prewitt) or second derivatives (Laplacian, Canny) to locate boundaries	Captures object outlines; works well with well-defined contrast	Produces fragmented edges; needs linking; sensitive to noise
Thresholding	Classifies pixels as foreground/background by comparing intensity to a threshold value (global or adaptive)	Very fast and simple; effective for high-contrast images	Fails with uneven illumination or low contrast between object and background
Region-based segmentation	Groups pixels into regions based on similarity criteria (region growing, splitting & merging)	Produces connected, coherent regions; robust to noise	Computationally more expensive; sensitive to seed-point selection

Practical Applications

- **Point detection:** isolated hot/dead pixel detection, satellite image anomaly spotting.
- **Line detection:** detecting cracks, wires, or road lane markings.
- **Edge detection:** object boundary extraction in medical imaging, industrial part outlining.
- **Thresholding:** document binarization, simple foreground/background separation with uniform lighting.
- **Region-based segmentation:** medical image organ segmentation, satellite land-cover classification.

Recommended Technique(s) for a Representative Scenario

(The original question references "this scenario" without a scenario description reaching this document, likely due to an accompanying figure/case study not captured in the extracted text. The recommendation below is illustrated using a representative industrial inspection scenario — separating a defective region from a mostly uniform product surface — and the same reasoning applies directly if you substitute your specific scenario.)

For separating a well-defined object or defect from a reasonably uniform background, a combination of thresholding (for fast, coarse foreground/background separation under controlled illumination) and edge detection (to precisely delineate the object's boundary once the region of interest is coarsely localized) is generally most appropriate. If the illumination is uneven or the object/background contrast is weak, region-based segmentation (region growing from a seed point within the object) is preferable, since it uses local similarity rather than a single global intensity threshold and is more robust to gradual illumination changes. Point and line detection would only be appropriate as supplementary techniques if the target features are literally isolated pixels or thin linear defects (e.g., hairline cracks) rather than extended regions.

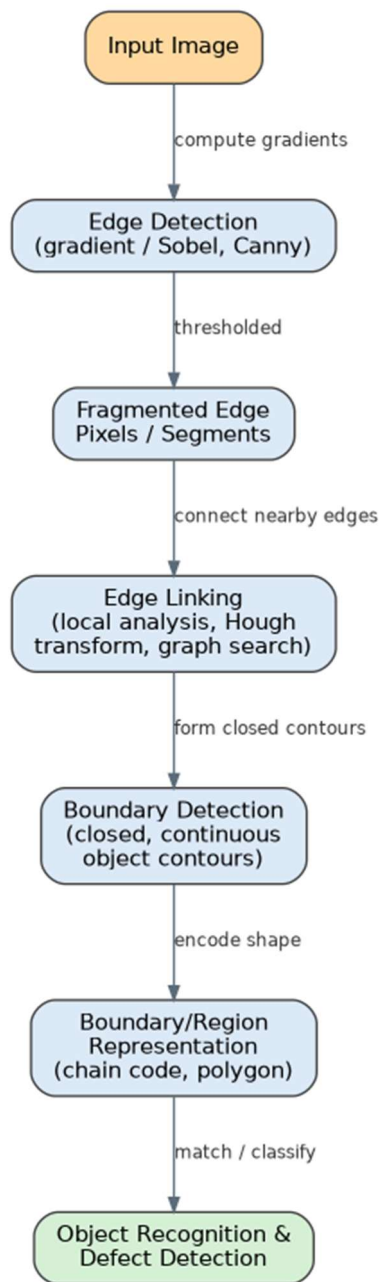
Q5. Edge Detection, Edge Linking, and Boundary Detection

Importance

Edge detection, edge linking, and boundary detection together form the pipeline that converts raw intensity-gradient information into usable object descriptions. Edge detection alone typically produces fragmented, discontinuous edge pixels due to noise, weak contrast, or occlusion; without linking and boundary formation, these fragments cannot reliably represent complete object shapes. This pipeline is therefore essential wherever accurate object shape, size, or contour information is required — for example, in object recognition, industrial defect detection, and medical image analysis.

Working Principles

- **Edge detection:** computes the local intensity gradient (using operators such as Sobel, Prewitt, or the multi-stage Canny detector) and marks pixels where the gradient magnitude exceeds a threshold, indicating a likely object boundary.
- **Edge linking:** connects individual edge pixels or short edge segments into longer, continuous contours. This can be done locally (analyzing gradient direction/magnitude similarity between neighboring edge pixels) or globally (using techniques such as the Hough transform to fit lines/curves across the whole image, or graph-based search methods).
- **Boundary detection:** organizes the linked edges into closed, coherent boundaries that fully enclose an object region, often represented compactly using chain codes, polygonal approximations, or spline curves for further shape analysis.



Contribution to Object Recognition and Defect Detection

- Accurate boundaries allow shape descriptors (perimeter, area, circularity, chain-code signatures) to be computed and matched against known object templates, forming the basis of many object-recognition pipelines.
- In industrial inspection, closed boundaries around anomalous regions (cracks, scratches, dents) allow precise measurement of defect size and shape, which is essential for automated pass/fail quality decisions.
- Without edge linking, fragmented edges could be mistaken for multiple small, unrelated defects; without boundary closure, region-based measurements (area, perimeter) cannot be computed reliably.

Illustrative Example

Consider inspecting a metal sheet for a surface crack: Sobel/Canny edge detection first marks the pixels along the crack's high-contrast edges, but noise and lighting variation leave small gaps in the detected crack outline. Edge linking closes these gaps using directional continuity, producing one continuous crack contour instead of several disconnected fragments. Boundary detection then encodes this contour, from which the system computes the crack's

length and width, enabling an automated system to accurately flag the part as defective — illustrating how the three stages progressively convert raw gradient data into an actionable engineering measurement.